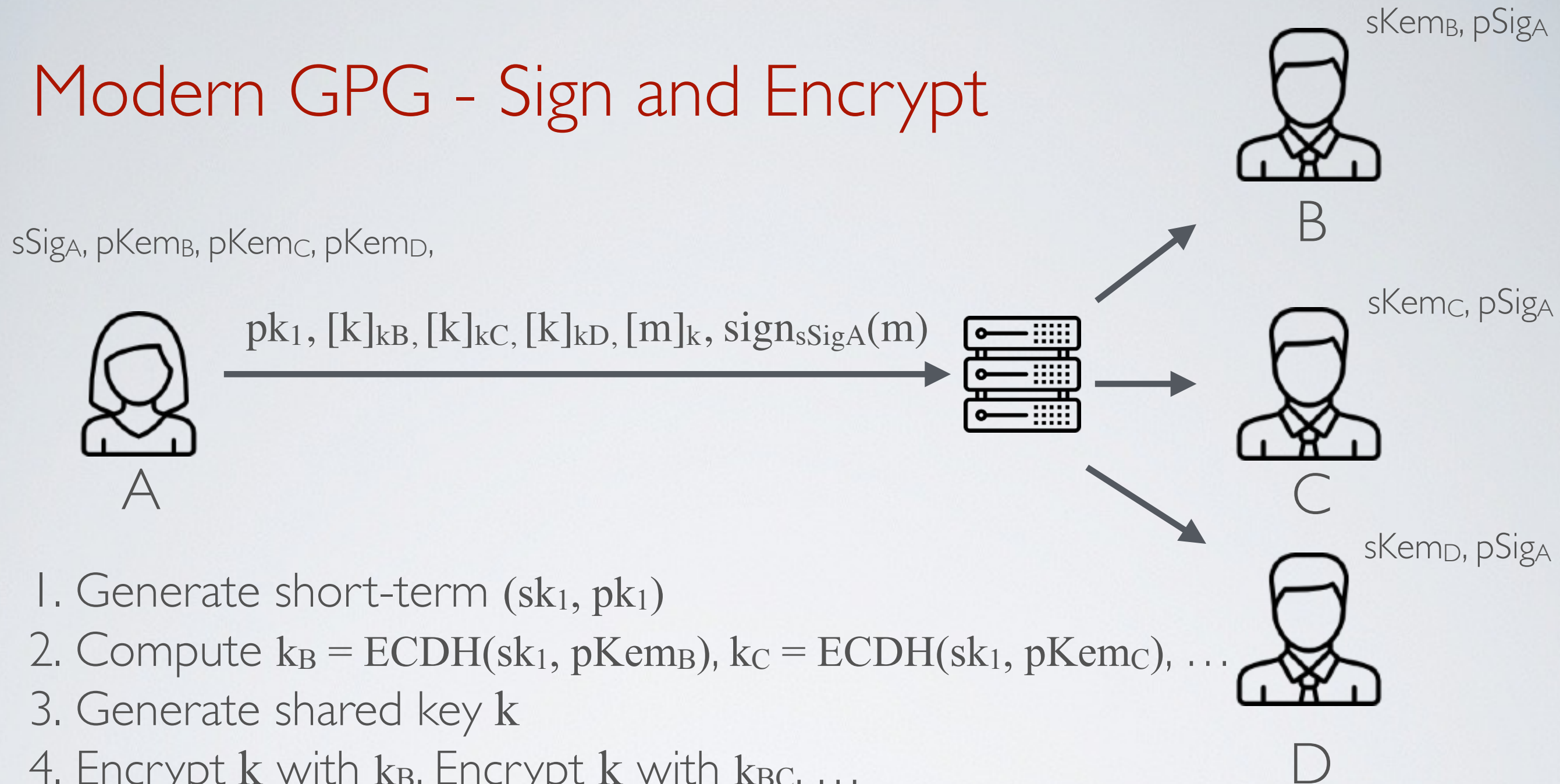


Modern GPG - Sign and Encrypt



1. Generate short-term (sk_1, pk_1)
2. Compute $k_B = \text{ECDH}(sk_1, pK_{em_B})$, $k_C = \text{ECDH}(sk_1, pK_{em_C})$, ...
3. Generate shared key k
4. Encrypt k with k_B , Encrypt k with k_{BC} , ...
5. Encrypt message m with k
6. Sign message m with $s\text{Sig}_A$

Limitations of GPG

- ◎ **Does not prevent Replay Attacks**

The same message can be repeated twice

- ◎ **Does not prevent Forward Secrecy**

If the long-term key for key exchange is compromised, the attacker can read previous message exchanged